

Multiple Choice (20 marks)

1. Which of the following is the main nitrogenous compound in urine?
 - (a) Uric acid
 - (b) Ammonia
 - (c) Urea
 - (d) Creatinine
2. What are the main metabolites of alcohol metabolism?
 - (a) Acetaldehyde, acetate,
 - (b) Pyruvate, beta-hydroxybutyrate,
 - (c) Acetyl-CoA, lipids,
 - (d) Acetone, lactate
3. What characteristic is not representative of a type IIb muscle fibre?
 - (a) Low oxidative capacity
 - (b) High fatigue resistance
 - (c) High motor unit strength
 - (d) Fast contractile speed
4. Which of the following statements about glucose metabolism is correct?
 - (a) All of the reactions of glycolysis are freely reversible for gluconeogenesis.
 - (b) Fructose cannot be used for gluconeogenesis in the liver.
 - (c) Red blood cells can catalyse aerobic glycolysis because they contain oxygen bound to haemoglobin.
 - (d) Red blood cells can only metabolise glucose by anaerobic glycolysis and the pentose phosphate pathway.
5. Which country has undergone the most rapid nutrition transition in the last two decades?
 - (a) China
 - (b) Sweden
 - (c) Britain
 - (d) India
6. Deficiency of which of the following vitamins results in megaloblastic anaemia?

- (a) Folate
 - (b) Niacin
 - (c) Riboflavin
 - (d) Vitamin B6
7. What is food security?
- (a) It relates to efforts to prevent terrorists from poisoning food supplies.
 - (b) It is about ensuring everyone's access to food.
 - (c) Its component elements include availability, utilisation, and stability, as well as access.
 - (d) Food security focuses primarily on ending micronutrient malnutrition.
8. Higher dietary sodium (salt) intake is generally associated with:
- (a) Decreased calcium excretion in the urine
 - (b) Increased risk of fracture
 - (c) Decreased dietary calcium absorption
 - (d) Negative calcium balance and bone mineral loss
9. Which of the following statements about human amino acid requirements in human nutrition is correct?
- (a) The amino acid requirements determined for adults is assumed to be the requirement pattern for all ages.
 - (b) The amino acid requirements determined for adults is assumed to be the requirements for maintenance for all ages.
 - (c) The amino acid pattern of breast milk is assumed to be the requirement pattern for all children.
 - (d) The amino acid pattern of human tissue protein is assumed to be the requirement patterns for infants and young children.
10. Which of the following changes in body composition are associated with ageing in older adults?
- (a) Increased bone mass
 - (b) Increased muscle mass
 - (c) Increased body fat
 - (d) All of the above

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which of the following foods might be considered a "goitrogen"?' is 'Fish'.
2. True or False: The correct answer to 'Eosinophilic oesophagitis' is 'May cause oesophageal obstruction'.
3. True or False: The correct answer to 'Pregnancy and lactation is known to:' is 'Decrease the rate of dietary calcium absorption'.
4. True or False: The correct answer to 'Deficiencies of which of the following nutrients can lead to anaemia?' is 'Copper and iron'.
5. True or False: The correct answer to 'Which of the following is the most plausible explanation for the protective effect of dietary fibre against cancer of the colon, as of 2020?' is 'Butyric acid, formed during colonic fibre fermentation stimulates anti-oxidant defences in the colon'.
6. True or False: The correct answer to 'Which one of the following is a definition of glycaemic index?' is 'The increase in blood insulin concentration after consuming the food compared with that after consuming an equivalent amount of white bread.'
7. True or False: The correct answer to 'A food additive is considered to be safe when:' is 'Its toxic effects are observed only at doses 100x the EDI'.
8. True or False: The correct answer to 'Since the 1996 World Food Summit, how has the number of food insecure people in the world changed?' is 'The number increased slightly.'
9. True or False: The correct answer to 'From the options listed, which type of carbohydrate cannot be metabolised by oral bacteria' is 'starch'.
10. True or False: The correct answer to 'Infants born to mothers who are vegan may be at increased risk of deficiency of which nutrient?' is 'Folate'.

End of Paper